

1. Administrative & Study Identification

**Full Study Title:** A Study to Investigate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe COPD (D5241C00006) **Summary Title:** A Multicenter, Randomized, Double-blind, Placebo-controlled, Parallel Group, Phase 3 Study to Evaluate the Efficacy and Safety of Tezepelumab in Adults with Moderate to Very Severe Chronic Obstructive Pulmonary Disease (COPD) **Short Title/Acronym:** EMBARK **Protocol Number:** D5241C00006 **Posting ID:** 448321960393699413 **NCT Number:** NCT06883305 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor Name:** AstraZeneca (Company: AstraZeneca) **Investigational Product:** Tezepelumab **Phase of Development:** Phase 3 (Phase ID: PHASE3, Criteria Type: PHASE\_REQUIREMENT) **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective** \* **Primary Objective:** To evaluate the efficacy of tezepelumab in reducing the annualized rate of moderate or severe COPD exacerbations (flare-ups) compared to a placebo. \* **Secondary Objectives:** To evaluate the effect of tezepelumab on lung function (change from baseline in FEV1), health-related quality of life, and to assess the overall safety and tolerability of the subcutaneous injections.

**2.2 Background & Rationale** To address the high burden of disease and frequent exacerbations in patients with moderate to very severe COPD who continue to experience flare-ups despite standard inhaled maintenance therapy. Type 2 inflammation is a known inflammatory endotype of COPD. Tezepelumab is a monoclonal antibody that targets thymic stromal lymphopoietin (TSLP), aiming to reduce airway inflammation and exacerbation frequency in this vulnerable patient population.

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3. Study Design & Methodology

**3.1 Design Framework** \* **Study Type:** Interventional \* **Control Type:** Placebo-controlled \* **Blinding:** Double-blind \* **Randomization:** Randomized (Parallel Group)

**3.2 Planned Enrollment & Duration** \* **Target \$N\$:** 990 participants \* **Number of Sites:** 275 global locations \* **Estimated Study Start:** 2024 \* **Estimated Primary Completion:** Late 2026/2027

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1.  $\geq 40$  to  $\leq 80$  years old. 2. Confirmed COPD diagnosis for  $\geq 1$  year prior to screening. 3. Post-BD FEV1  $\geq 20\%$  and  $\leq 70\%$  PN, FEV1/FVC  $< 0.70$  at screening. 4. Triple (ICS+LABA+LAMA) or dual inhaled COPD therapy, if triple therapy is considered not appropriate, for  $\geq 3$  consecutive months prior to V1. 5.  $\geq 2$  moderate or  $\geq 1$  severe COPD exacerbations in the prior year; At least 1 of 2 moderate exacerbations must require the use of systemic corticosteroids. At least one of the previous exacerbations should be confirmed to have occurred while the participant was on triple or dual inhaled maintenance therapy. 6. EOS  $\geq 150$  cells/ $\mu$ L during screening. 7. CAT  $\geq 15$  at screening. 8. Former or current smokers with a history of smoking  $\geq 10$  pack-years.

**4.2 Exclusion Criteria** 1. Clinically important pulmonary disease or radiological findings suggestive of a respiratory disease other than COPD. 2. Asthma, incl. pediatric, or asthma-COPD overlap syndrome (ACOS). 3. Any unstable disorder that can impact participant safety or study outcomes. 4. Tuberculosis requiring treatment within 12 months prior V2. 5. Malignancies current or past. 6. LTOT  $> 4.0$  L/min or O2 saturation  $< 89\%$  despite LTOT.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen** \* **Investigational Product:** Tezepelumab (Preferred Name: Tezepelumab / Drug Name: tezepelumab) \* **Trade Name:** Tezspire \* **ATC Code:** R03DX11 \* **Dosage/Frequency:** Monthly subcutaneous (SC) injections of one of two different doses of tezepelumab, or placebo. \* **Route of Administration:** Subcutaneous (SC) injection via pre-filled syringe. \* **Duration of Treatment:** Maximum treatment duration of 76 weeks and a minimum of 52 weeks, plus an off-treatment safety follow-up period of 12 weeks.

**5.2 Concomitant & Prohibited Therapy** \* **Permitted:** Standard background COPD inhaled maintenance therapy (e.g., ICS, LABA, LAMA). \* **Prohibited:** \* Macrolides (less than 6 months). \* Systemic immuno-suppressive or immuno-modulating medications during the study period.

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## 6. Study Timeline & Visit Schedule

| Visit ID  | Window (Days) | Key Activities/Procedures                                                          |
|-----------|---------------|------------------------------------------------------------------------------------|
| Screening | 2 to 4 weeks  | Informed Consent, Medical History, Baseline Lung Function (FEV1)                   |
| Baseline  | Day 0         | Randomization, First SC Injection, Baseline Questionnaires                         |
| Interim   | Every 4 Weeks | SC Injections, Lung function monitoring (FEV1), Exacerbation tracking, Safety Labs |
| End of    |               |                                                                                    |

**Study** | Week 52-76 + 12 Weeks | Final lung function measurements, Questionnaire completion, 12-week safety follow-up |

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## **7. Outcome Measures (Endpoints)**

**7.1 Primary Endpoint \* Definition:** Annualised rate of moderate or severe COPD exacerbations: The annualised moderate or severe COPD exacerbation rate (based on exacerbations reported by the investigator) up to 76 weeks treatment period compared to placebo.

**7.2 Secondary & Exploratory Endpoints** \* Change from baseline in post-bronchodilator (post-BD) forced expiratory volume in 1 second (FEV1). \* Change from baseline in pre-dose/pre-bronchodilator (pre-BD) forced expiratory volume in 1 second (FEV1). \* Change from baseline in the St. George's Respiratory Questionnaire (SGRQ) total score. \* Frequency and severity of treatment-emergent adverse events (TEAEs).

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## **8. Safety & Adverse Event Reporting**

- **Adverse Event (AE) Monitoring:** Patient reporting of symptoms/exacerbations via questionnaires, and regular blood tests to measure medication levels and check for antibodies.
  - **Serious Adverse Events (SAEs):** Reported within 24 hours of site/investigator awareness.
  - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee utilized to oversee safety data throughout the trial.
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## **9. Statistical Analysis Plan (SAP) Summary**

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy endpoints; Safety Set for AE/SAE evaluations.
  - **Sample Size Justification:** \$N=990\$, powered adequately to detect a statistically significant reduction in the annualized COPD exacerbation rate.
  - **Handling of Missing Data:** Multiple Imputation (MI) or standard generalized linear models (e.g., negative binomial regression for exacerbation rates) accounting for patient follow-up time.
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## **10. Quality Assurance & Regulatory Compliance**

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

- **Monitoring Plan:** Regular onsite visits for source data verification combined with remote data monitoring.
  - **Audit Trail:** Electronic Case Report Form (eCRF) locking and comprehensive data clarification form tracking.
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## 11. Study Identifiers & Governance

- **Primary ID:** D5241C00006
  - **Registry Number:** NCT06883305
  - **Posting ID:** 448321960393699413
  - **Sponsor/Lead Organization:** AstraZeneca
  - **Study Title:** EMBARK
  - **Official Regulatory Title:** A Randomised, Double-blind, Placebo-controlled, Parallel Group, Multicentre, Phase III Study to Evaluate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe Chronic Obstructive Pulmonary Disease
  - **Trial Status:** RECRUITING
  - **Source Level:** 5
  - **EMA URL:** [Tezspire EPAR Product Information](#)
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## 12. Clinical Framework (COPD Specific)

- **Condition Name:** COPD (Chronic Obstructive Pulmonary Disease)
  - **Disease Areas:** Chronic Obstructive Pulmonary Disease (COPD)
  - **Preferred Name:** Chronic Obstructive Airway Disease
  - **Semantic Type:** Disease or Syndrome
  - **Definition:** A disease of chronic diffuse irreversible airflow obstruction. Subcategories of COPD include CHRONIC BRONCHITIS and PULMONARY EMPHYSEMA.
  - **Ontology Codes:**
    - MeSH Code: D029424
    - HPO Code: HP:0006510
    - SNOMED ID: 13645005
  - **Optimization Target:** Reduction in the annualized rate of COPD exacerbations and improvement in lung function
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## 13. Study Procedures & Methodology

- **Management Pathway:** Standard maintenance inhaler therapy augmented with monthly SC biologic injections
  - **Education Protocol (HCP):** Investigator training on standardized COPD exacerbation definitions and reporting.
  - **Education Protocol (Patient):** Symptom monitoring, exacerbation recognition, and safety reporting instructions.
  - **Quality Audit Mechanism:** Continuous eCRF monitoring and scheduled spirometry quality reviews.
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## 14. Observation & Follow-Up Schedule

- **Total Duration:** Maximum 76 weeks of treatment (minimum 52 weeks) + 12 weeks of safety follow-up
- **Onsite Visit Cadence:** Every 4 weeks (monthly) for SC injections and evaluations
- **Remote Monitoring Frequency:** Between-visit symptom tracking and reporting as needed.
- **Checklist Review Interval:** Routine efficacy and safety laboratory draws per protocol schedule.

## 15. Sign-off & Approval

| Role                     | Name   | Signature | Date          | : | --- | : | --- | : | --- | : | --- |
|--------------------------|--------|-----------|---------------|---|-----|---|-----|---|-----|---|-----|
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] | : | --- | : | --- | : | --- | : | --- |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] | : | --- | : | --- | : | --- | : | --- |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] | : | --- | : | --- | : | --- | : | --- |